In[ ]:=
```
datalambda = {366, 405, 436, 578, 546, 405}; (*nm*)
```

In[ ]:=
```
dataV0 = {1.472, 1.181, 1.035, 0.417, 0.521}; (*V*)
```

In[ ]:=
```
data = {{819.1050765, 1.472}, {740.2282914, 1.181},
   {687.5973807, 1.035}, {518.6720727, 0.417}, {549.0704359, 0.521}}; (*THz,V*)
```

In[ ]:=
```
fit = NonlinearModelFit[data, A * x - B, {A, B}, x]
      ajusta a modelo no lineal
```

Out[ ]=

```
FittedModel[ -1.4 + 0.00351 x ]
```

In[ ]:=
```
fit[{"BestFit", "ParameterTable"}]
```

Out[ ]=

$\{-1.40113 + 0.00350914\, x,$

	Estimate	Standard Error	t-Statistic	P-Value
A	0.00350914	0.0000644093	54.4819	0.0000136203
B	1.40113	0.0433237	32.341	0.0000649704

$\}$

In[ ]:=
```
plotfit =
  Plot[fit[x], {x, 400, 1000}, BaseStyle → {FontFamily → "CambriaMath", FontSize → 14},
  representación gráfica          estilo base     familia de tipo de letra    tamaño de tipo de letra
   PlotStyle → {Blend[{Blue, White}], Thick}, PlotRange → {{500, 850}, {0.35, 1.55}},
   estilo de repre··· mezcla··· azul  blanco   grueso   rango de representación
   Frame → True, FrameLabel → {Style["ν / THz", Bold, 16], Style["V₀ / eV", 16, Bold]}];
   marco   verd···  etiqueta de marco  estilo   negrita   estilo   negrita
```

In[ ]:=
```
plotdata = ListPlot[data, PlotStyle → {Black, Thickness[0.003]}];
           representación de ·· estilo de repre··· negro  grosor
```

In[ ]:=
```
(*errores*)
```

In[ ]:=
```
errnu[A_] := ((3 * 10^8 / ((datalambda⟦A⟧ * 10^(-9))^2)) * 10^(-9)) / (10^12);
```

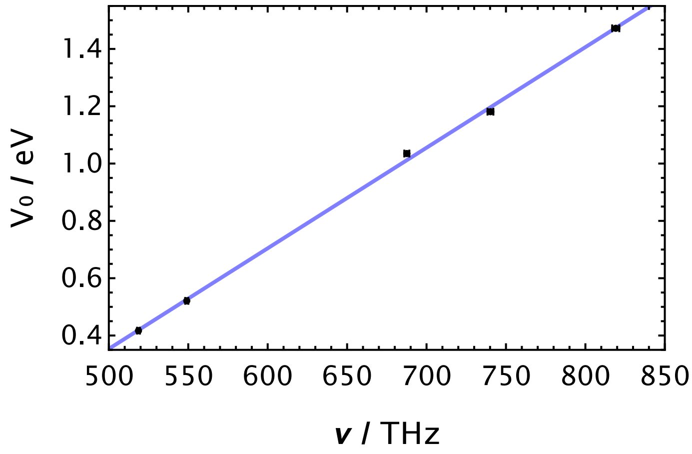
In[ ]:=
```
Needs["ErrorBarPlots`"]
necesita
```

General: ErrorBarPlots` is now obsolete. The legacy version being loaded may conflict with current functionality. See the Compatibility Guide for updating information.

In[ ]:=
```
errV0[B_] := (0.8 / 100) * dataV0[B] + 5 * 10^(-3);
```

In[ ]:=
```
dataerr = ErrorListPlot[{{{819.1050765, 1.472}, ErrorBar[errnu[1], errV0[1]]},
     {{740.2282914, 1.181}, ErrorBar[errnu[2], errV0[2]]},
     {{687.5973807, 1.035}, ErrorBar[errnu[3], errV0[3]]},
     {{518.6720727, 0.417}, ErrorBar[errnu[4], errV0[4]]},
     {{549.0704359, 0.521}, ErrorBar[errnu[5], errV0[5]]}},
   AxesOrigin → {0, 0}, PlotStyle → {Black, Thickness[0.003]}];
   origen de ejes        estilo de repre··· negro  grosor
```

In[ ]:=
```
Show[plotfit, dataerr]
muestra
```
Out[ ]=



In[ ]:=
```
h = 0.0035091437335052103 * (1.6 * 10^(-19)) * (10^(-12))
```
Out[ ]=

5.61463×10^{-34}

In[ ]:=
```
errh = 0.000064409301899442 01 * (1.6 * 10^(-19)) * (10^(-12))
```
Out[ ]=

1.03055×10^{-35}

In[ ]:=
```
hpdg = 2 * Pi * (1.054571817 * 10^(-34))
             número pi
```
Out[ ]=

6.62607×10^{-34}

In[ ]:=
```
errabs = Norm[h - hpdg]
         norma
```
Out[ ]=

1.01144×10^{-34}

In[ ]:=
```
errrel = errabs / hpdg
```
Out[ ]=

0.152646

In[ ]:=
```
(*-----------------------------------------------------------------------*)
```
In[ ]:=
```
(*CON EL APARATO DE LA CTE DE PLANCK*)
```
In[ ]:=
```
datalambda2 = {525, 505, 588, 611, 472} * 10^(-9); (*en m*)
```
In[ ]:=
```
datanu2 = N[299792458 / datalambda2] * 10^(-12); (*en THz*)
          valor numérico
```
In[ ]:=
```
dataW2 = {0.419, 0.475, 0.149, 0.081, 0.650}; (*en eV*)
```
In[ ]:=
```
data2 = Transpose[{datanu2, dataW2}];
        transposición
```

In[ ]:=
```
fit2 = NonlinearModelFit[data2, A * x - B, {A, B}, x]
```
ajusta a modelo no lineal

Out[ ]=
```
FittedModel[ -1.86 + 0.00395 x ]
```

In[ ]:=
```
fit2[{"BestFit", "ParameterTable"}]
```

Out[ ]=

$\{-1.85965 + 0.00395389\, x,$

	Estimate	Standard Error	t-Statistic	P-Value
A	0.00395389	0.000123258	32.0781	0.0000665772
B	1.85965	0.0693453	26.8172	0.000113778

$\}$

In[ ]:=
```
plotfit2 =
  Plot[fit2[x], {x, 400, 1000}, BaseStyle → {FontFamily → "CambriaMath", FontSize → 14},
   PlotStyle → {Blend[{Blue, White}], Thick}, PlotRange → {{460, 650}, {0, 0.68}},
   Frame → True, FrameLabel → {Style["ν / THz", Bold, 16], Style["V₀ / eV", 16, Bold]}];
```
representación gráfica, estilo base, familia de tipo de letra, tamaño de tipo de letra, estilo de repre···, mezcla···, azul, blanco, grueso, rango de representación, marco, verd···, etiqueta de marco, estilo, negrita, estilo, negrita

In[ ]:=
```
(*errores*)
```

In[ ]:=
```
errnu2[A_] := ((3 * 10^8 / ((datalambda2[[A]])^2)) * 10^(-9)) / (10^12);
```

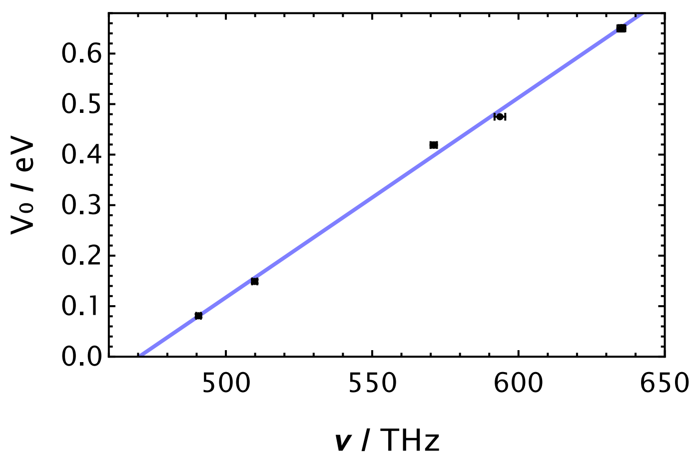
In[ ]:=
```
errV02[B_] := (0.5 / 100) * dataW2[[B]] + 3 * 10^(-3);
```

In[ ]:=
```
dataerr2 = ErrorListPlot[{{{datanu2[[1]], dataW2[[1]]}, ErrorBar[errnu2[1], errV02[1]]},
     {{datanu2[[2]], dataW2[[2]]}, ErrorBar[errnu[2], errV0[2]]},
     {{datanu2[[3]], dataW2[[3]]}, ErrorBar[errnu2[3], errV02[3]]},
     {{datanu2[[4]], dataW2[[4]]}, ErrorBar[errnu2[4], errV02[4]]},
     {{datanu2[[5]], dataW2[[5]]}, ErrorBar[errnu2[5], errV02[5]]}},
   AxesOrigin → {0, 0}, PlotStyle → {Black, Thickness[0.003]}];
```
origen de ejes, estilo de repre···, negro, grosor

In[ ]:=
```
plotdata2 = ListPlot[data2, PlotStyle → {Black, Thickness[0.003]}];
```
representación de li···, estilo de repre···, negro, grosor

In[ ]:=
```
Show[plotfit2, dataerr2]
```
muestra

Out[ ]=



In[ ]:=
```
h2 = 0.003953889097949258 * (1.6 * 10^(-19)) * (10^(-12))
```

Out[ ]=

6.32622×10^{-34}

In[ ]:=
```
errh2 = 0.00012325801595869084 * (1.6 * 10^(-19)) * (10^(-12))
```

Out[ ]=

1.97213×10^{-35}

In[ ]:=
```
errrel2 = Norm[h2 - hpdg] / hpdg
          norma
```

Out[ ]=

0.0452527